

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

A salesperson’s total earnings consist of a base salary of x dollars per year, plus commission earnings of **11%** of the total sales the salesperson makes during the year. This year, the salesperson has a goal for the total earnings to be at least **3** times and at most **4** times the base salary. Which of the following inequalities represents all possible values of total sales s , in dollars, the salesperson can make this year in order to meet that goal?

Answer

- A. $2x \leq s \leq 3x$
- B. $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$
- C. $3x \leq s \leq 4x$
- D. $\frac{3}{0.11}x \leq s \leq \frac{4}{0.11}x$

Correct Answer: B

Rationale

Choice B is correct. It’s given that a salesperson's total earnings consist of a base salary of x dollars per year plus commission earnings of **11%** of the total sales the salesperson makes during the year. If the salesperson makes s dollars in total sales this year, the salesperson’s total earnings can be represented by the expression $x + 0.11s$. It’s also given that the salesperson has a goal for the total earnings to be at least **3** times and at most **4** times the base salary, which can be represented by the expressions $3x$ and $4x$, respectively. Therefore, this situation can be represented by the inequality $3x \leq x + 0.11s \leq 4x$. Subtracting x from each part of this inequality yields $2x \leq 0.11s \leq 3x$. Dividing each part of this inequality by **0.11** yields $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$. Therefore, the inequality $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$ represents all possible values of total sales s , in dollars, the salesperson can make this year in order to meet their goal.

Choice A is incorrect. This inequality represents a situation in which the total sales, rather than the total earnings, are at least **2** times and at most **3** times, rather than at least **3** times and at most **4** times, the base salary.

Choice C is incorrect. This inequality represents a situation in which the total sales, rather than the total earnings, are at least **3** times and at most **4** times the base salary.

Choice D is incorrect. This inequality represents a situation in which the total earnings are at least **4** times and at most **5** times, rather than at least **3** times and at most **4** times, the base salary.